
Reducing Traffic Congestion in Hanoi, Vietnam Using Artificial Intelligence and Internet of Things

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Abstract – Hanoi is one of the largest cities in Vietnam. Traffic congestion is one of the most serious urban transportation problems in Hanoi, Vietnam. Rapid urbanization, increasing private vehicle ownership, limited road capacity, frequent traffic violations, and highly variable traffic demand contribute to long travel times, fuel consumption, air pollution, and reduced quality of life. Traditional traffic management systems based on fixed-time traffic signals and manual monitoring are often unable to respond quickly to changing traffic conditions. Artificial Intelligence (AI) and the Internet of Things (IoT) provide an opportunity to develop a more adaptive and intelligent transportation system. This paper proposes an AI- and IoT-based traffic management solution for Hanoi. The proposed system combines IoT sensors, AI-enabled cameras, GPS/mobile data, edge computing, cloud platforms, adaptive traffic signals, and an intelligent traffic control center. Traffic information is collected continuously and analyzed using computer vision, machine learning, and reinforcement learning. Based on predicted traffic conditions, traffic signals can automatically adjust green, yellow, and red-light durations and coordinate neighboring intersections to create green-wave corridors. The system can also detect incidents, illegal parking, traffic violations, and abnormal congestion and provide real-time information to road users. The proposed approach is supported by recent research on AI, IoT, adaptive traffic signal control, and intelligent transportation systems. Importantly, Hanoi has already begun deploying AI cameras and intelligent traffic control infrastructure. Recent operational results reported by Hanoi indicate substantial improvements in travel time and intersection throughput. The paper therefore combines AI and IoT technologies to significantly reduce the traffic congestion in Hanoi, Vietnam.

Keywords: Internet of Things, AI, Traffic congestion management

I. INTRODUCTION

Hanoi is the capital of Vietnam and one of the country's fastest-growing metropolitan areas. The city has experienced rapid economic development, urban expansion, and growth in travel demand. At the same time, the number of private cars and motorcycles has increased significantly. These factors place considerable pressure on the existing road network. Traffic congestion occurs particularly during morning and afternoon peak periods, at major intersections, around commercial areas, near schools and universities, and on important radial and ring-road corridors. Congestion is not caused by a single factor. It results from the interaction between traffic demand, road capacity, traffic-signal operation, vehicle behavior, parking activity, incidents, construction, weather, and traffic violations. Fig.1 illustrates an example of traffic congestion in Hanoi, Vietnam.



Figure 1. An example of traffic congestion in Hanoi, Vietnam

Traditional traffic control generally depends on predetermined signal schedules. Such systems work reasonably well when traffic patterns are predictable, but they become less effective when traffic demand changes suddenly. For example, an intersection may receive a large number of vehicles from one direction while the programmed signal continues to allocate the same amount of green time to all approaches. The result can be long queues in one direction and unused capacity in another. Intelligent Transportation Systems (ITS) attempt to overcome this limitation by connecting vehicles, roads, sensors, cameras, traffic signals, communication networks, and control centers. IoT provides the communication and sensing infrastructure, while AI provides the intelligence required to interpret the collected data and make decisions.

Research has demonstrated the potential of IoT-based adaptive traffic management systems. An IoT traffic-management architecture can collect information from vehicles and infrastructure and transmit it to intelligent decision-making systems [1]. More recent research has investigated deep reinforcement learning (DRL), in which an AI agent learns traffic-signal policies according to traffic conditions [2], [3]. Reviews of DRL-based traffic signal control indicate that adaptive AI control can potentially reduce delay and improve traffic flow, although transferring models from simulation to real-world networks remains a major challenge [4].

The Hanoi context is particularly interesting because the city has already moved toward AI-enabled traffic management. Hanoi's intelligent traffic-control program includes AI cameras, traffic monitoring, violation detection, and adaptive traffic-signal control. The city's intelligent transportation plan also identifies functions

including traffic monitoring, traffic control, public transport management, parking management, incident management, infrastructure management, and traffic simulation [5].

In January 2026, Hanoi reported that its AI camera system had been operating with approximately 1,837 cameras at 195 major intersections and 25 major streets. The system can collect traffic information and support real-time signal control. Reported results included travel-time reductions of approximately 31–36.8% on one-way roads and 20–25% on heavily trafficked two-way roads, together with increases in intersection throughput [6].

Therefore, the main objective of this paper is to develop a practical framework for reducing traffic congestion in Hanoi through the integration of AI and IoT. The proposed system builds on existing infrastructure while introducing predictive analytics, adaptive signal optimization, edge computing, multimodal transportation data, and coordinated corridor management.

II. RELATED WORKS

Traffic congestion has become a major challenge in rapidly urbanizing cities such as Hanoi, where high traffic demand, heterogeneous vehicle flows, frequent congestion at intersections, and limited road capacity make conventional fixed-time traffic signal control increasingly ineffective. Early intelligent traffic-management research focused mainly on adaptive control, allowing signal timings to respond to changing traffic conditions rather than relying on predetermined schedules. Abdulhai et al. [7] investigated reinforcement learning for adaptive traffic signal control and demonstrated the potential of Q-learning to dynamically select signal-control actions according to the observed traffic environment. Their work established an important foundation for subsequent AI-based traffic signal control because the controller could learn an appropriate policy instead of relying entirely on predefined traffic models.

For the Vietnamese context, Phan et al. [8] proposed an IoT-based intelligent traffic-light system combined with the green-wave theory. The system collected traffic conditions from multiple intersections through an IoT platform and centrally controlled traffic signals. Programmable logic controllers were deployed at intersections to collect traffic parameters and communicate them to a central operation system. The approach demonstrated how IoT can provide the real-time data infrastructure required for intelligent traffic management in Vietnamese urban environments. However, the control strategy was primarily based on green-wave optimization rather than a learning-based AI model, limiting its ability to adapt autonomously to highly dynamic traffic conditions.

Fuzzy logic was subsequently investigated as an alternative for dealing with the uncertainty and variability of urban traffic. Nguyen et al. [9] developed a fuzzy traffic signal control method for a single intersection under mixed traffic conditions and evaluated it using a typical intersection in Hanoi. The method dynamically adjusted the green-light duration according to traffic conditions and achieved lower average delay than the existing signal-control strategy in simulation. This study is particularly relevant to Hanoi because Vietnamese traffic consists of a heterogeneous mixture of motorcycles, cars, buses, and trucks, making conventional traffic models less suitable for local conditions.

Vuong et al. [10] further developed an adaptive neuro-fuzzy inference system (ANFIS) specifically for an isolated intersection in Hanoi. Their method combined neural-network learning with fuzzy inference to determine appropriate green times according to fluctuating traffic volumes. The proposed system was evaluated using VISSIM and MATLAB and demonstrated better adaptability and traffic performance than fixed-time and conventional fuzzy-control methods. This study represents an important transition from rule-based fuzzy control toward data-driven AI-based traffic signal optimization for Hanoi. Nevertheless, the system focused on an isolated intersection, while congestion in Hanoi is often caused by interactions among multiple neighboring intersections.

As deep learning became increasingly integrated with reinforcement learning, DRL emerged as a promising solution for complex traffic-control problems. Rasheed et al. [11] provided a comprehensive review of deep reinforcement learning for traffic signal control, showing that DRL can learn control policies directly from high-dimensional traffic states and overcome some limitations of conventional model-based approaches. The review also identified multi-intersection coordination, training complexity, simulation-to-real-world transfer, and the availability of reliable traffic data as important challenges. These issues are particularly relevant to Hanoi, where a practical AI traffic-control system must operate under highly dynamic and heterogeneous traffic conditions.

Haddad et al. [12] extended DRL toward cooperative control of multiple intersections. Their method formulated traffic signal control as a multi-agent reinforcement learning problem in which individual agents controlled different intersections while exchanging information about neighboring traffic states, actions, and rewards. The cooperative approach improved average waiting time, queue length, and CO₂ emissions compared with several existing methods, particularly under congested conditions. Such multi-agent coordination is potentially suitable for Hanoi because congestion frequently propagates between adjacent intersections rather than occurring independently at a single junction.

Ren et al. [13] subsequently proposed a two-layer coordinated reinforcement-learning framework for traffic networks. The method combined local cooperation among neighboring intersections with global coordination across the network, while simultaneously considering traffic congestion and vehicle emissions. Experiments using synthetic networks and the Cologne urban network demonstrated the potential of coordinated multi-agent learning for large-scale traffic management. This approach suggests a possible direction for Hanoi in which local intersection controllers can cooperate while receiving information from a city-level traffic-management platform.

More recent research has focused on improving the robustness and convergence of DRL-based traffic signal control. Cai and Wei [14] proposed an enhanced DRL method incorporating double Q-learning, a dueling network, prioritized sampling, and noise parameters. Their approach was designed to improve training efficiency and robustness under different traffic-flow conditions, and experiments showed improvements in queue length and waiting time. Such methods are promising for Hanoi because traffic patterns can change substantially between peak and off-peak periods and because a traffic-control model must remain effective under different demand conditions.

The research trend has also shifted toward integrating multiple AI techniques with adaptive traffic signal control. Agrahari et al. [15] reviewed fuzzy logic, reinforcement learning, deep reinforcement learning, metaheuristic optimization, and hybrid approaches for adaptive traffic signal control. Their review emphasized that modern systems need to move beyond isolated intersections toward coordinated multi-intersection environments and real-time traffic information. This development is particularly important for Hanoi, where an effective congestion-reduction system should combine real-time sensing, network-level coordination, and adaptive decision-making rather than optimizing individual intersections independently.

Finally, recent studies have increasingly emphasized the convergence of AI, IoT, and predictive analytics for intelligent urban mobility. Gheorghe and Soica [16] reviewed AI- and IoT-enabled adaptive traffic-control systems and highlighted the role of IoT sensors in continuously collecting traffic information, while AI models process these data to support real-time signal optimization, traffic prediction, vehicle coordination, and emergency response. This integrated architecture is highly relevant to Hanoi because IoT can provide the sensing and communication infrastructure while AI can transform real-time traffic data into adaptive control decisions.

III. PROPOSED SOLUTIONS

This section presents the solutions to reduce the traffic congestion using AI and IoT technologies.

AI solutions to reduce the traffic congestion

Artificial Intelligence (AI) provides an effective approach for reducing urban traffic congestion by enabling traffic-management systems to learn from historical and real-time traffic data and make adaptive decisions. Unlike conventional traffic-control systems that rely on predefined signal timings, AI-based approaches can identify complex traffic patterns and dynamically adjust traffic-control strategies according to current traffic conditions. For Hanoi, where traffic demand varies significantly between peak and off-peak periods and where motorcycles, cars, buses, and trucks share the same road network, AI can provide more flexible traffic-management capabilities.

One important AI solution is machine-learning-based traffic prediction. Machine-learning models such as Random Forest, Support Vector Regression, XGBoost, LightGBM, and Artificial Neural Networks can be trained using historical traffic data to predict traffic volume, vehicle speed, travel time, and congestion levels. By predicting congestion before it becomes severe, traffic-management systems can proactively modify signal timings, recommend alternative routes, or provide warnings to road users. Time-series models such as Long Short-Term

Memory (LSTM) networks are particularly suitable because traffic conditions exhibit strong temporal dependencies. An LSTM model can learn traffic patterns from previous time intervals and predict future traffic conditions for individual roads or intersections.

Another important AI solution is intelligent traffic signal control. Reinforcement learning (RL) and Deep Reinforcement Learning (DRL) allow traffic signal controllers to learn appropriate actions from interactions with the traffic environment. The traffic state can include queue length, vehicle density, waiting time, and traffic flow, while the actions can include changing the signal phase or extending and reducing green-light duration. The objective function can be designed to minimize vehicle waiting time, queue length, travel time, or a combination of these factors. DRL methods are particularly promising because deep neural networks can represent complex relationships between traffic states and control actions.

AI can also be used for traffic-flow optimization and route recommendation. A central traffic-management system can analyze traffic conditions across multiple roads and recommend routes that reduce congestion. Graph Neural Networks (GNNs) are increasingly suitable for this task because a road network can naturally be represented as a graph, where intersections are nodes and roads are edges. GNN-based models can capture spatial relationships between neighboring roads and intersections, while recurrent or attention-based components can model temporal traffic variations. This enables congestion prediction to consider both the current traffic condition and congestion propagation from neighboring locations.

Computer vision represents another important AI solution. Cameras installed at intersections can provide video streams that are processed by object-detection models such as YOLO, Faster R-CNN, or other lightweight deep-learning networks. These models can detect and classify vehicles, estimate traffic density, measure queue length, identify stopped vehicles, and detect incidents such as accidents or abnormal traffic behavior. Compared with manually collected traffic information, AI-based video analysis can continuously provide detailed traffic information without requiring physical sensors for every vehicle.

For Hanoi, AI solutions should also consider heterogeneous traffic conditions. The large proportion of motorcycles creates traffic-flow characteristics that differ from those of cities dominated by passenger cars. Therefore, AI models should be trained using local traffic data and should explicitly consider different vehicle classes. A practical AI-based congestion-reduction system could combine traffic prediction, computer vision, reinforcement learning, and route optimization. Historical and real-time data could first be used to predict congestion, after which an AI controller could optimize traffic signals and recommend alternative routes.

Despite their potential, AI-only solutions have an important limitation: they require reliable and continuous data. An AI model cannot make accurate decisions if traffic information is incomplete, delayed, or inaccurate. Consequently, AI should be supported by an appropriate sensing and communication infrastructure. This limitation motivates the use of IoT technologies to continuously collect traffic data from roads and intersections and transmit these data to AI-based traffic-management systems.

IoT solutions to reduce the traffic congestion

The Internet of Things (IoT) provides the sensing, communication, and data-exchange infrastructure required for intelligent traffic management. An IoT-based transportation system connects physical devices, sensors, vehicles, traffic lights, cameras, and communication networks to continuously collect and exchange information. In contrast to traditional traffic-management systems that depend on periodic manual surveys or fixed signal plans, IoT enables traffic conditions to be monitored in near real time. This capability is particularly useful for Hanoi, where traffic conditions can change rapidly because of high vehicle density, road incidents, weather conditions, and variations in travel demand.

One important IoT solution is the deployment of traffic sensors at intersections and along major roads. Sensors can measure vehicle volume, speed, occupancy, queue length, and other traffic parameters. Different sensing technologies can be used depending on the road environment and cost requirements, including inductive loops, infrared sensors, ultrasonic sensors, radar sensors, Bluetooth/Wi-Fi detectors, and video cameras. The collected information can be transmitted through IoT communication networks to a central traffic-management platform. This provides traffic authorities with a continuous overview of traffic conditions across the road network.

IoT-enabled smart traffic lights provide another effective solution. Conventional traffic lights operate according to predefined schedules and cannot respond efficiently to sudden changes in traffic demand. IoT-enabled traffic signals can receive traffic information from connected sensors and dynamically modify signal timings. For example, when sensors detect a long queue on one approach of an intersection, the system can increase the corresponding green-light duration. Similarly, if an approach has very low traffic volume, its green time can be reduced. Multiple intersections can also exchange information through an IoT network to support coordinated traffic-signal operation.

IoT can also support connected vehicles and vehicle-to-infrastructure communication. Vehicles equipped with communication devices can transmit information such as position, speed, direction, and travel status to roadside infrastructure. Traffic-management centers can aggregate these data to obtain a more comprehensive understanding of traffic conditions. In addition, traffic authorities can send information back to connected vehicles, including congestion warnings, recommended routes, accident notifications, and information about traffic-signal changes. Such communication can improve both traffic efficiency and road safety.

Another IoT application is real-time incident detection. IoT devices can monitor roads continuously and identify abnormal conditions such as accidents, stalled vehicles, road blockages, or unusually slow traffic. Once an incident is detected, the information can be transmitted to a traffic-management center. Authorities can then respond more rapidly by dispatching emergency services, modifying traffic signals, or providing alternative-route information. Faster incident detection and response can prevent local disturbances from developing into large-scale congestion.

IoT infrastructure can additionally support smart parking and traffic-demand management. Parking sensors can determine the availability of parking spaces and transmit the information to a centralized platform. Drivers can then receive information about available parking locations, reducing the amount of time spent searching for parking. In dense urban areas such as Hanoi, this can reduce unnecessary vehicle circulation and consequently reduce local traffic pressure.

However, an IoT-only traffic-management system mainly focuses on data collection and communication. Although IoT can provide large volumes of real-time information, it does not inherently provide advanced reasoning or optimization capabilities. Large amounts of sensor data must be analyzed to determine appropriate traffic-control actions. Therefore, IoT infrastructure should be combined with AI algorithms capable of predicting traffic conditions, detecting patterns, and optimizing traffic-control decisions. The integration of these two technologies can provide a more intelligent and autonomous traffic-management system.

Combination of AI and IoT solutions to reduce the traffic congestion

The integration of Artificial Intelligence (AI) and the Internet of Things (IoT) provides a comprehensive approach to intelligent traffic management. IoT provides the sensing and communication infrastructure for continuously collecting traffic information, while AI analyzes these data and determines appropriate traffic-management actions. Therefore, the combination of AI and IoT can transform a conventional traffic-control system into an adaptive, predictive, and potentially autonomous transportation system. This approach is particularly suitable for Hanoi because the city's traffic conditions are highly dynamic and involve heterogeneous vehicle flows.

A typical AI-IoT traffic-management architecture can be divided into four main layers: the sensing layer, communication layer, intelligence layer, and application/control layer. At the sensing layer, IoT devices such as cameras, radar sensors, inductive-loop detectors, GPS devices, and connected vehicles collect traffic information. Typical parameters include vehicle volume, speed, density, queue length, vehicle type, and location. At the communication layer, these data are transmitted through wireless and wired communication technologies to edge devices or a central traffic-management platform. At the intelligence layer, AI models analyze the collected information to predict traffic conditions, detect abnormal events, and optimize traffic-control decisions. Finally, the application/control layer applies the AI decisions to traffic lights, route recommendations, variable message signs, and other transportation services.

One important application is AI-based adaptive traffic signal control using IoT data. IoT sensors continuously provide traffic states at intersections, while an AI controller uses these states to determine the optimal signal

phases. For example, a reinforcement-learning agent can receive queue lengths and traffic volumes from multiple approaches as its state. It can then select actions such as extending the current green phase, changing the signal phase, or coordinating the signal with neighboring intersections. The reward function can be designed to minimize total waiting time, queue length, travel time, and unnecessary stops. This enables traffic signals to adapt automatically to real-time traffic conditions.

A second application is real-time congestion prediction. IoT devices continuously generate traffic data that can be stored in a traffic database. AI models such as LSTM, GRU, XGBoost, LightGBM, or Graph Neural Networks can use historical and real-time observations to predict traffic conditions several minutes into the future. If the model predicts that congestion will occur at a particular intersection, the traffic-management system can proactively modify signal timings or recommend alternative routes before severe congestion develops. This changes traffic management from a reactive approach to a predictive approach.

The integration of AI and IoT also enables coordinated multi-intersection control. Instead of optimizing each intersection independently, IoT devices can exchange traffic information among neighboring intersections. AI-based multi-agent reinforcement learning can then coordinate traffic signals across the network. Each intersection can be represented by an intelligent agent that observes local traffic conditions while receiving information from neighboring agents. This approach can reduce the propagation of queues from one intersection to another and improve traffic flow along major corridors.

Computer vision and IoT can also be combined for automated traffic monitoring. Cameras installed at intersections can transmit video streams to edge-computing devices, where AI models detect and classify vehicles. Edge AI can reduce communication latency by processing video close to the data source rather than transmitting all raw video to a central server. The extracted information, such as vehicle counts, queue lengths, and traffic density, can then be transmitted to the central traffic-management system. This architecture can reduce bandwidth requirements while enabling near-real-time traffic control.

For Hanoi, a practical AI-IoT system could integrate roadside cameras, traffic sensors, GPS information from public and private vehicles, weather information, and traffic-signal controllers. IoT devices would continuously collect traffic data across major roads and intersections. AI models would process these data to estimate the current traffic state and predict future congestion. A reinforcement-learning or optimization module could subsequently determine appropriate signal timings and route-control strategies. The system could also provide congestion information to drivers through mobile applications or variable message signs.

The major advantage of the AI-IoT approach is the closed-loop operation of sensing, communication, prediction, decision-making, and control. Traffic conditions are first sensed by IoT devices, transmitted through the communication infrastructure, analyzed by AI models, and converted into control actions. The resulting traffic conditions are then sensed again, allowing the system to continuously update its decisions. Such a closed-loop architecture can provide greater adaptability than either AI or IoT alone.

Nevertheless, several challenges must be addressed before large-scale deployment in Hanoi. These include sensor reliability, communication latency, data quality, cybersecurity, privacy, computational requirements, and the need for large-scale local traffic datasets. In addition, AI models trained in other cities may not perform reliably under Hanoi's heterogeneous traffic conditions. Therefore, the proposed system should be trained and evaluated using real traffic data collected from Hanoi and should be designed to support incremental deployment. Overall, the integration of AI and IoT provides a promising foundation for developing a scalable smart-traffic system capable of reducing congestion, improving travel efficiency, and supporting intelligent transportation management in Hanoi.

IV. THE DEVELOPMENT OF THE SOLUTIONS IN HANOI, VIETNAM

Hanoi has gradually shifted from conventional traffic-management methods toward intelligent transportation systems (ITS), with increasing use of sensors, cameras, communication technologies, artificial intelligence (AI), and real-time traffic control. In the early stages, traffic congestion was mainly addressed through infrastructure expansion, conventional traffic signals, manual traffic control, and fixed-time signal plans. These approaches were relatively simple to operate but had limited ability to respond to rapidly changing traffic demand. The increasing number of vehicles and the heterogeneous traffic composition of Hanoi, particularly the large proportion of motorcycles, created a need for more adaptive traffic-management technologies.

The development of intelligent traffic solutions initially focused on collecting traffic information and improving coordination between intersections. IoT technologies provided an important foundation by enabling traffic sensors, cameras, signal controllers, and communication devices to exchange information. In particular, IoT-based traffic-light systems and green-wave strategies have been investigated for Vietnamese urban environments. These systems demonstrated the feasibility of using real-time traffic information to coordinate traffic signals and improve vehicle movement along major corridors. Research conducted specifically for Hanoi subsequently investigated fuzzy logic and adaptive neuro-fuzzy inference systems (ANFIS) for traffic signal control, demonstrating the potential of intelligent controllers to adjust signal timing according to changing traffic conditions.

Hanoi has subsequently moved toward a city-scale smart-traffic strategy. In 2024, the city approved its Smart Transportation Project for the 2024--2030 period, with a vision toward 2045. The project planned the development of an integrated intelligent traffic operation center with functions including traffic monitoring, traffic-information provision, traffic control, public transportation management, parking management, incident management, and traffic-infrastructure management. The initial phase also included the deployment of intelligent transportation-system equipment at major intersections and roads. Later phases planned to expand intelligent traffic infrastructure to cover a much larger proportion of the city's road network.

A major development occurred in 2025 with the deployment of AI-based traffic cameras and real-time intelligent traffic signal control. In December 2025, Hanoi officially launched an upgraded traffic-control center integrating AI-based traffic management. The system connected 1,837 cameras with 188 AI-enabled traffic-signal control cabinets. In addition, 957 cameras were installed at 195 signalized intersections to provide real-time traffic-flow information. The system was designed not only to detect traffic violations but also to identify congestion, establish green-wave corridors, and dynamically shorten or extend signal phases according to traffic demand.

The initial results indicate the practical potential of this AI-based approach. Six green-wave corridors were established on major streets, and the Hue, Hang Bai (Hanoi) corridor demonstrated a 31% reduction in travel time compared with the previous traffic-control system. The system also reduced the number of stops at intersections, shortened queues at red lights, and improved traffic flow through intersections. After approximately one month of operation, Hanoi reported improvements in traffic throughput and travel time on several monitored corridors, with some one-way corridors achieving travel-time reductions of approximately 31-36.8% and increases in intersection throughput of approximately 13--18.88%.

The development continued in 2026 with the expansion and refinement of AI-based traffic monitoring. A six-month pilot using AI cameras on Vo Chi Cong and Pham Van Dong (Hanoi) streets demonstrated the ability to collect traffic-flow information, count vehicles by lane, monitor lane violations, and support traffic analysis. In July 2026, Hanoi further introduced electronic traffic-information displays and an AI virtual assistant for traffic management. The system was reported to operate with nearly 5,000 AI cameras and could analyze and summarize real-time traffic information to support traffic-control decisions.



Figure 2. Example of the application of AI solution to reduce traffic congestion in Hanoi, Vietnam.

V. CONCLUSION

Traffic congestion is a complex problem that cannot be solved by road construction alone. Hanoi requires a transportation system capable of continuously observing traffic, predicting future conditions, and adapting to changing demand. This paper proposed an integrated AI and IoT solution consisting of AI cameras, IoT sensors, GPS/mobile data, edge computing, traffic prediction, adaptive traffic-signal control, green-wave coordination, incident detection, intelligent parking, and centralized traffic management.

The next step should therefore be to move from individual AI applications toward a fully integrated, predictive, network-wide intelligent transportation system. Such a system should combine real-time IoT data, AI forecasting, reinforcement learning, digital twins, connected vehicles, public transportation, and human supervision. If implemented carefully, AI and IoT can help Hanoi reduce congestion, shorten travel time, improve road safety, reduce emissions, improve emergency response, and provide a foundation for a smarter and more sustainable urban transportation system.

Conflicts of Interest

The authors declare that there is no conflict of interest concerning the publishing of this paper.

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